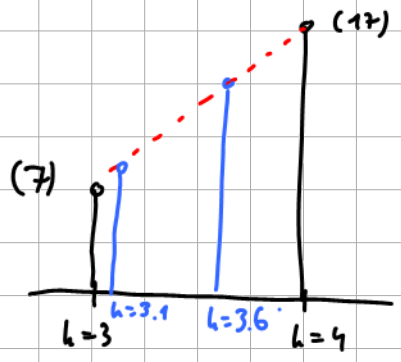
1)



Linear interpolation

$$X(h) = m \cdot h + n$$

$$m = \frac{17-7}{4-3} = 10$$

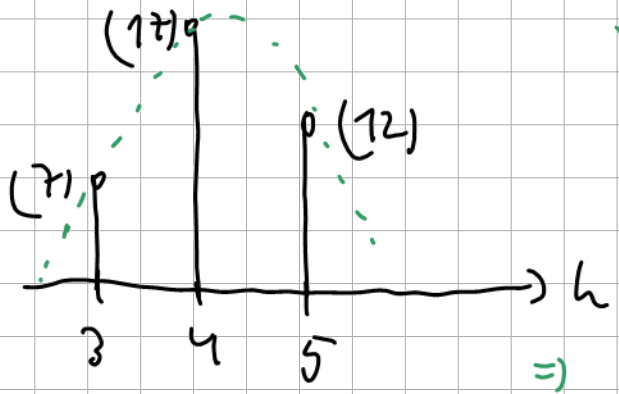
$$7 = 10 \cdot 3 + n \iff n = 7 - 30 = -23$$

$$\Rightarrow X(h) = 10 \cdot h - 23$$

$$X(h=3.1) = 10 \cdot 3.1 - 23 = 31 - 23 = 8$$

$$X(h=3.6) = 10 \cdot 3.6 - 23 = 36 - 23 = 13$$

2)



$$X(h) = a h^2 + b h + c$$

$$7 = a \cdot 9 + b \cdot 3 + c$$
$$17 = a \cdot 16 + b \cdot 4 + c$$
$$12 = a \cdot 25 + b \cdot 5 + c$$

$$\Rightarrow \quad 10 = 7a + b \qquad \cdot (-2)$$
$$\Rightarrow \quad 5 = 16a + 2b \qquad +$$

$$\Rightarrow \quad -15 = 2a$$
$$\Rightarrow \quad a = -7{,}5$$

$$b = 10 - 7 \cdot (-7{,}5) = 62{,}5$$

$$c = 7 - 9 \cdot (-7{,}5) - 3 \cdot 62{,}5 = -113$$

$$X_{(h=3{,}1)} = -7{,}5 \cdot 3{,}1^2 + 62{,}5 \cdot 3{,}1 - 113 = 8{,}68$$

$$X_{(h=3{,}6)} = -7{,}5 \cdot 3{,}6^2 + 62{,}5 \cdot 3{,}1 - 113 = 14{,}8$$

3) Yes, human voice stays understandable for different ranges of time stretching. Doubling the speed (TimeStretchFactor=0.5) and halving the speed (TimeStretchFactor=2.0) should be no real problem for intelligibility (understanding of voice).

4) Yes, changing the pitch in the range betweet 0.6<PitchShiftFactor<1.5 should be no problem regarding the intelligibility.

5) Increasing the amount of available data is always good for training more generalised Neural Networks. Dataset augmentation is a simple way of increasing the amount of available data for more stable training.

In the case of multimedia data, dataset augmentation teaches the Neural Network, what kind of modifications in the input data is acceptable for human users.

6) Dataset augmentation for image classification tasks:
- Flipping left to right
- Changing coloured images into gray images
- Lowpassfiltering / Resampling of images
- Whitening or blackening of boxes inside the images (such a modification simulates data loss during transmission over the net)
-Inserting background noise
...

7) Dataset augmentation for keyword spotting

- Inserting 20 ms silence at random position (corresponds to a transmission loss)
- pitch shifting
- time stretching
- adding white Gaussian noise
- adding noise below the masking threshold
- inserting clipping
- ...

8) $f = 440\ Hz$

$$b = 6 \cdot \operatorname{arcsinh}\left(\frac{f}{660\ Hz}\right)$$

$$= 4.08$$

$$m = b \cdot 100 = 408\ mel$$

9) Frequency resolution of human ear: $\frac{1}{27}$ Bark

$$\left.\begin{aligned} 0 \text{ Bark} &\hat{=} 600 \cdot \sinh\left(\frac{0}{6}\right) = 0 \text{ Hz} \\ \frac{1}{27} \text{ Bark} &\hat{=} 600 \cdot \sinh\left(\frac{1/27}{6}\right) = 3.7 \text{ Hz} \end{aligned}\right\} \text{Frequency resolution} \approx 3.7 \text{ Hz}$$

$$\left.\begin{aligned} 24 \text{ Bark} &\hat{=} 600 \cdot \sinh\left(\frac{24}{6}\right) = 16374 \text{ Hz} \\ 24\tfrac{1}{27} \text{ Bark} &\hat{=} 600 \cdot \sinh\left(\frac{24 + \frac{1}{27}}{6}\right) = 16475 \text{ Hz} \end{aligned}\right\} \text{Frequency resolution} \approx 101 \text{ Hz}$$